

Answer Key to HW 6

1a) $y[n] = \left(\frac{(-2)^{n+1} - (e^{-1})^{n+1}}{-2 - e^{-1}} \right) u[n].$

1b) $y[n] = \frac{1}{2} \left(e^{-n} + \frac{(-2)^{n+1} - (e^{-1})^{n+1}}{2 + e^{-1}} \right) u[n]$

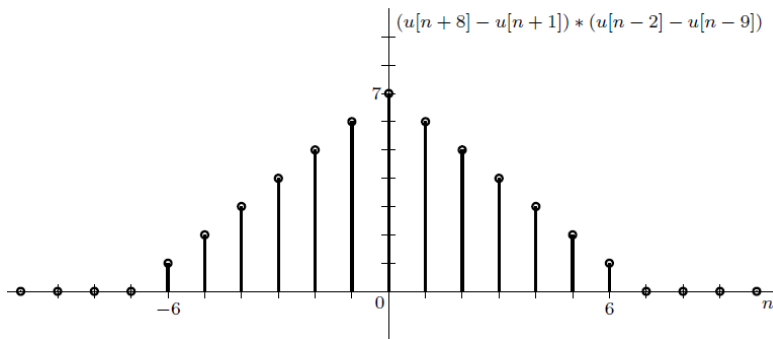
1c) $y[n] = \frac{18}{25} \left(\left(\frac{1}{3} \right)^n + 5n(2)^n - (2)^n \right) u[n]$

1d) $y[n] = 0.38 \left[3^{n+1} \cos\left[\frac{\pi}{3}(n+1) - 2.26\right] + 0.64(2)^{n+1} \right] u[n]$

2a) $h_p[n] = 2u[n] + (0.5)^n u[n] - u[n-5].$

2b) $h_c[n] = (n+3 - 2(0.5)^{n+1})u[n] - (n-2 - 2(0.5)^{n-4})u[n-5].$

3)



4a) $H_a(z) = \frac{2z^2 - z + 3}{z^2 - 2z + 1}$ 4b) $H_b(z) = \frac{z - 3}{z^2 + 0.5z - 0.8}$

5a) $H_a(z) = 1 + 4z^{-2} - 2z^{-4}$ 5b) $H_b(z) = \frac{z}{z - \gamma}$

6)

- (a) the system is both asymptotically and BIBO stable.
- (b) the system is marginally stable and BIBO unstable.
- (c) the system is both asymptotically and BIBO unstable.
- (d) the system is both asymptotically and BIBO unstable.
- (e) the system is marginally stable and BIBO unstable.

7)

Time constant, rise time, and pulse dispersion are all the same value $W_h = 3$. An input containing $(0.5)^n$ will cause system resonance.